Time Series Forecasting Appendix

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NON-SEASONAL TIME SERIES DATA

Step 1: Importing and Inspecting Data

```
# Import the necessary libraries
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(tseries)
library(ggplot2)
library(readr)

# Import the dataset
unrate_data <- read.csv("~/Desktop/MA641/UNRATE.csv")

# Convert Date column to Date type
unrate_data$observation_date <- as.Date(unrate_data$observation_date, format = "%Y-%m-%d")

# Convert the 'UNRATE' column to numeric
unrate_data$UNRATE <- as.numeric(unrate_data$UNRATE)

# Check for missing values and replace them with 0
unrate_data$UNRATE[is.na(unrate_data$UNRATE)] <- 0

# Create the unemployment data into a time series object
unemployment_ts <- ts(unrate_data$UNRATE, frequency = 12, start = c(1948, 1))

# Check the structure of the time series object
str(unemployment_ts)
```

```
##  Time-Series [1:710] from 1948 to 2007: 3.8 3.7 3.7 3.6 3.8 3.9 3.8 3.8 3.8 3.8 ...
```

Step 2: ADF Test for Stationarity

```
# Perform Augmented Dickey-Fuller test to check stationarity

check_stationarity <- function(series, alpha = 0.05) {
test_result <- adf.test(series)
p_val <- test_result$p.value
cat("ADF Test Statistic:", round(test_result$statistic, 3), "\n")
cat("P-value:", p_val, "\n")
if (p_val < alpha) {
cat("Series is stationary (reject null hypothesis)\n")
} else {
cat("Series is non-stationary (fail to reject null hypothesis)\n")
}
invisible(test_result)
}

# Running ADF Test on original data
cat("Running ADF Test on original data:\n")
```

```
## Running ADF Test on original data:
```

```
check_stationarity(unemployment_ts)
```

```
## ADF Test Statistic: -3.262
## P-value: 0.07727813
## Series is non-stationary (fail to reject null hypothesis)
```

Step 3: Data Transformation

```
# Convert to numeric and handle missing values
unemployment_ts <- as.numeric(unemployment_ts)
unemployment_ts[is.na(unemployment_ts)] <- 0
# Create time series object
unemployment_ts <- ts(unemployment_ts, frequency = 12, start = c(1948, 1))
# Check the structure and first few rows of the data
str(unrate_data)
```

```
## 'data.frame':    710 obs. of  2 variables:
##  $ observation_date: Date, format: "1966-08-01" "1966-09-01" ...
##  $ UNRATE          : num  3.8 3.7 3.7 3.6 3.8 3.9 3.8 3.8 3.8 3.8 ...
```

```
head(unrate_data)
```

```
##   observation_date UNRATE
## 1       1966-08-01    3.8
## 2       1966-09-01    3.7
```

```
## 3       1966-10-01    3.7
## 4       1966-11-01    3.6
## 5       1966-12-01    3.8
## 6       1967-01-01    3.9
```

Step 4: Fit AR, MA, and ARMA Models

```r
# Fit a Pure AR(1) model
fit_ar <- Arima(unemployment_ts, order = c(1, 0, 0))
summary(fit_ar)
```

```
## Series: unemployment_ts
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##          ar1    mean
##       0.9677  5.7823
## s.e.  0.0093  0.4967
##
## sigma^2 = 0.1977:  log likelihood = -432.28
## AIC=870.55   AICc=870.59   BIC=884.25
##
## Training set error measures:
##                       ME      RMSE       MAE        MPE     MAPE      MASE
## Training set 0.004781907 0.4439526 0.1587839 -0.3093286 2.566664 0.1785009
##                     ACF1
## Training set 0.04893059
```

```r
# Fit a Pure MA(1) model
fit_ma <- Arima(unemployment_ts, order = c(0, 0, 1))
summary(fit_ma)
```

```
## Series: unemployment_ts
## ARIMA(0,0,1) with non-zero mean
##
## Coefficients:
##          ma1    mean
##       0.8714  5.9218
## s.e.  0.0132  0.0712
##
## sigma^2 = 1.033:  log likelihood = -1018.75
## AIC=2043.5   AICc=2043.53   BIC=2057.2
##
## Training set error measures:
##                        ME    RMSE       MAE       MPE     MAPE      MASE
## Training set 0.0008575093 1.01503 0.7752518 -4.580428 13.88198 0.8715184
##                    ACF1
## Training set 0.7096683
```

```r
# Fit an ARMA(1,1) model
fit_arma <- Arima(unemployment_ts, order = c(1, 0, 1))
summary(fit_arma)
```

```
## Series: unemployment_ts
## ARIMA(1,0,1) with non-zero mean
##
## Coefficients:
##          ar1     ma1    mean
##       0.9636  0.0616  5.8121
## s.e.  0.0102  0.0420  0.4693
##
## sigma^2 = 0.1973:  log likelihood = -431.21
## AIC=870.42   AICc=870.47   BIC=888.68
##
## Training set error measures:
##                       ME     RMSE       MAE        MPE     MAPE      MASE
## Training set 0.003995652 0.443284 0.1589857 -0.3318038 2.574163 0.1787277
##                      ACF1
## Training set -0.004215036
```

Step 5: Fit ARIMA Models

```r
# Fit an ARIMA(1,1,1) model
fit_arima_1_1_1 <- Arima(unemployment_ts, order = c(1, 1, 1))
summary(fit_arima_1_1_1)
```

```
## Series: unemployment_ts
## ARIMA(1,1,1)
##
## Coefficients:
##           ar1     ma1
##       -0.7272  0.7962
## s.e.   0.1210  0.1054
##
## sigma^2 = 0.1989:  log likelihood = -432.5
## AIC=871   AICc=871.04   BIC=884.69
##
## Training set error measures:
##                       ME      RMSE       MAE         MPE     MAPE      MASE
## Training set 0.000825423 0.4450153 0.1529708 -0.09810348 2.422708 0.1719659
##                     ACF1
## Training set -0.02517534
```

```r
# Fit an ARIMA(0,1,1) model
fit_arima_0_1_1 <- Arima(unemployment_ts, order = c(0, 1, 1))
summary(fit_arima_0_1_1)
```

```
## Series: unemployment_ts
```

```
## ARIMA(0,1,1)
##
## Coefficients:
##          ma1
##       0.0408
## s.e.  0.0413
##
## sigma^2 = 0.2003:  log likelihood = -435.59
## AIC=875.19   AICc=875.21   BIC=884.32
##
## Training set error measures:
##                       ME      RMSE       MAE        MPE     MAPE      MASE
## Training set 0.000822734 0.4469694 0.1502242 -0.0985417 2.374515 0.1688782
##                      ACF1
## Training set -0.003520204
```

```
# Fit an ARIMA(2,1,2) model
fit_arima_2_1_2 <- Arima(unemployment_ts, order = c(2, 1, 2))
summary(fit_arima_2_1_2)
```

```
## Series: unemployment_ts
## ARIMA(2,1,2)
##
## Coefficients:
##          ar1     ar2      ma1      ma2
##       0.0632  0.4660  -0.0266  -0.5744
## s.e.  0.2294  0.2134   0.2138   0.2006
##
## sigma^2 = 0.1983:  log likelihood = -430.53
## AIC=871.07   AICc=871.15   BIC=893.89
##
## Training set error measures:
##                        ME      RMSE       MAE        MPE     MAPE      MASE
## Training set 0.0009419562 0.4437764 0.1572726 -0.1323584 2.499542 0.1768019
##                      ACF1
## Training set 0.002581726
```

Step 6: Model Comparison (AIC and BIC)

```
# Compare models based on AIC and BIC
AIC(fit_ar, fit_ma, fit_arma, fit_arima_1_1_1, fit_arima_0_1_1, fit_arima_2_1_2)
```

```
## Warning in AIC.default(fit_ar, fit_ma, fit_arma, fit_arima_1_1_1,
## fit_arima_0_1_1, : models are not all fitted to the same number of observations
```

```
##                 df       AIC
## fit_ar           3  870.5549
## fit_ma           3 2043.5008
## fit_arma         4  870.4181
## fit_arima_1_1_1  3  871.0024
```

```
## fit_arima_0_1_1  2  875.1898
## fit_arima_2_1_2  5  871.0664
```

```
BIC(fit_ar, fit_ma, fit_arma, fit_arima_1_1_1, fit_arima_0_1_1, fit_arima_2_1_2)
```

```
## Warning in BIC.default(fit_ar, fit_ma, fit_arma, fit_arima_1_1_1,
## fit_arima_0_1_1, : models are not all fitted to the same number of observations
```

```
##                 df       BIC
## fit_ar           3  884.2507
## fit_ma           3 2057.1966
## fit_arma         4  888.6792
## fit_arima_1_1_1  3  884.6940
## fit_arima_0_1_1  2  884.3176
## fit_arima_2_1_2  5  893.8857
```

Step 7: Box-Ljung Test for Residuals

```
# Perform the Ljung-Box test to check for autocorrelation in residuals
Box.test(fit_ar$residuals, lag = 12, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_ar$residuals
## X-squared = 9.9353, df = 12, p-value = 0.6216
```

```
Box.test(fit_ma$residuals, lag = 12, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_ma$residuals
## X-squared = 4131.7, df = 12, p-value < 2.2e-16
```

```
Box.test(fit_arma$residuals, lag = 12, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_arma$residuals
## X-squared = 8.4655, df = 12, p-value = 0.7478
```

```
Box.test(fit_arima_1_1_1$residuals, lag = 12, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_arima_1_1_1$residuals
## X-squared = 4.7534, df = 12, p-value = 0.9657
```

```r
Box.test(fit_arima_0_1_1$residuals, lag = 12, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_arima_0_1_1$residuals
## X-squared = 10.253, df = 12, p-value = 0.5938
```

```r
Box.test(fit_arima_2_1_2$residuals, lag = 12, type = "Ljung-Box")
```

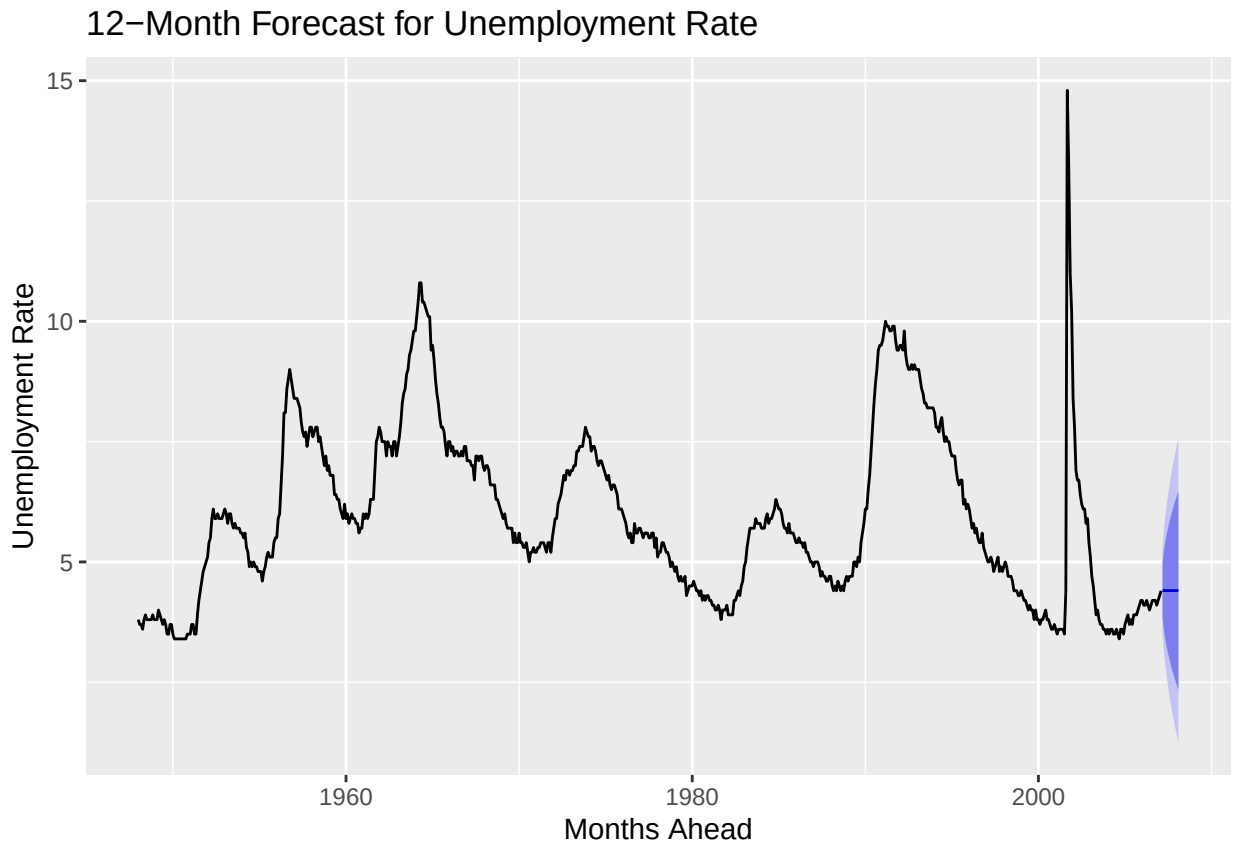
```
##
##  Box-Ljung test
##
## data:  fit_arima_2_1_2$residuals
## X-squared = 1.4793, df = 12, p-value = 0.9999
```

```r
fit_arima_0_1_1 <- Arima(unemployment_ts, order = c(0, 1, 1))
summary(fit_arima_0_1_1)
```

```
## Series: unemployment_ts
## ARIMA(0,1,1)
##
## Coefficients:
##          ma1
##       0.0408
## s.e.  0.0413
##
## sigma^2 = 0.2003:  log likelihood = -435.59
## AIC=875.19   AICc=875.21   BIC=884.32
##
## Training set error measures:
##                       ME      RMSE       MAE        MPE     MAPE      MASE
## Training set 0.000822734 0.4469694 0.1502242 -0.0985417 2.374515 0.1688782
##                      ACF1
## Training set -0.003520204
```

```r
### Step 9: Forecasting the next 12 months
# Forecast using the ARIMA model (fit_arima) for the next 12 months
unrate_forecast <- forecast(fit_arima_0_1_1, h = 12)

# Plot the forecast
autoplot(unrate_forecast) +
  ggtitle("12-Month Forecast for Unemployment Rate") +
  xlab("Months Ahead") +
  ylab("Unemployment Rate")
```



Step 9: Final Conclusion :

In this analysis, I explored different AR, MA, ARMA, and ARIMA models to forecast the Unemployment Rate based on historical data. By comparing models using AIC and BIC values, the ARIMA(0,1,1) model provided the best fit for the data. I performed the Ljung-Box test to confirm that the residuals of the model were white noise, indicating that the chosen model adequately captured the structure of the data. Finally, used the best-fitting ARIMA model to forecast future values of the Unemployment Rate, demonstrating the utility of time series forecasting for economic indicators.

SEASONAL TIME SERIES DATA

Step 1: Importing the dataset

```
cpi_data <- read.csv("~/Desktop/MA641/cpiai.csv")
head(cpi_data)
```

```
##         Date Index Inflation
## 1 1913-01-01   9.8        NA
## 2 1913-02-01   9.8      0.00
## 3 1913-03-01   9.8      0.00
## 4 1913-04-01   9.8      0.00
## 5 1913-05-01   9.7     -1.02
## 6 1913-06-01   9.8      1.03
```

```r
colnames(cpi_data) <- c("Date", "Index", "CPIAI")

# Convert 'Date' column to Date format
cpi_data$Date <- as.Date(cpi_data$Date, format = "%m/%d/%Y")

# Handle missing values
cpi_data$CPIAI[is.na(cpi_data$CPIAI)] <- 0  # Replace NA with 0

# Ensure 'CPIAI' is numeric
cpi_data$CPIAI <- as.numeric(cpi_data$CPIAI)

# Remove duplicate rows
cpi_data <- cpi_data[!duplicated(cpi_data), ]
```

Step 2: Checking for Stationarity using ADF Test

```r
# Perform the ADF test
adf_test <- adf.test(cpi_data$CPIAI, alternative = "stationary")
```

```
## Warning in adf.test(cpi_data$CPIAI, alternative = "stationary"): p-value
## smaller than printed p-value
```

```r
# Output the result
cat("ADF Test Statistic:", adf_test$statistic, "\n")
```

```
## ADF Test Statistic: -5.960888
```

```r
cat("P-value:", adf_test$p.value, "\n")
```

```
## P-value: 0.01
```

```r
# Check if the series is stationary
if (adf_test$p.value < 0.05) {
  cat("Series is stationary (reject null hypothesis)\n")
} else {
  cat("Series is non-stationary (fail to reject null hypothesis)\n")
}
```

```
## Series is stationary (reject null hypothesis)
```

```r
# If the series is non-stationary, difference the data
if (adf_test$p.value >= 0.05) {
  cpi_data_diff <- diff(cpi_data$CPIAI, differences = 1)
  cat("Differenced data to achieve stationarity\n")
}
```

Fitting Models:

```r
# Fit ARIMA models if enough data
    fit_ar <- Arima(cpi_data$CPIAI, order = c(1, 0, 0))  # Adjust based on your data
    summary(fit_ar)
```

```
## Series: cpi_data$CPIAI
## ARIMA(1,0,0) with non-zero mean
##
## Coefficients:
##          ar1    mean
##       0.4916  0.3214
## s.e.  0.0267  0.0354
##
## sigma^2 = 0.3441:  log likelihood = -936.94
## AIC=1879.89   AICc=1879.91   BIC=1894.78
##
## Training set error measures:
##                       ME      RMSE       MAE MPE MAPE      MASE       ACF1
## Training set 0.0001129641 0.5860646 0.3767845 NaN  Inf 0.8939275 -0.0753164
```

```r
    fit_ma <- Arima(cpi_data$CPIAI, order = c(0, 0, 1))  # Adjust based on your data
    summary(fit_ma)
```

```
## Series: cpi_data$CPIAI
## ARIMA(0,0,1) with non-zero mean
##
## Coefficients:
##          ma1    mean
##       0.3719  0.3220
## s.e.  0.0235  0.0258
##
## sigma^2 = 0.3741:  log likelihood = -981.04
## AIC=1968.08   AICc=1968.11   BIC=1982.98
##
## Training set error measures:
##                        ME      RMSE       MAE MPE MAPE      MASE      ACF1
## Training set -6.374969e-05 0.6110215 0.3932252 NaN  Inf 0.9329332 0.1011928
```

```r
    fit_arma <- Arima(cpi_data$CPIAI, order = c(1, 0, 1))  # Adjust based on your data
    summary(fit_arma)
```

```
## Series: cpi_data$CPIAI
## ARIMA(1,0,1) with non-zero mean
##
## Coefficients:
##          ar1      ma1    mean
##       0.8847  -0.5899  0.3181
## s.e.  0.0268   0.0501  0.0618
##
## sigma^2 = 0.3241:  log likelihood = -904.83
## AIC=1817.67   AICc=1817.71   BIC=1837.53
##
```

```
## Training set error measures:
##                       ME      RMSE      MAE MPE MAPE      MASE      ACF1
## Training set 0.0007770963 0.5685077 0.366862 NaN  Inf 0.8703862 0.0588272
```

```
    # Fit a SARIMA model (adjust the parameters as needed)
    fit_sarima <- Arima(cpi_data$CPIAI, order = c(1, 1, 1), seasonal = c(1, 1, 1))

    # Check the summary of the fitted SARIMA model
    summary(fit_sarima)
```

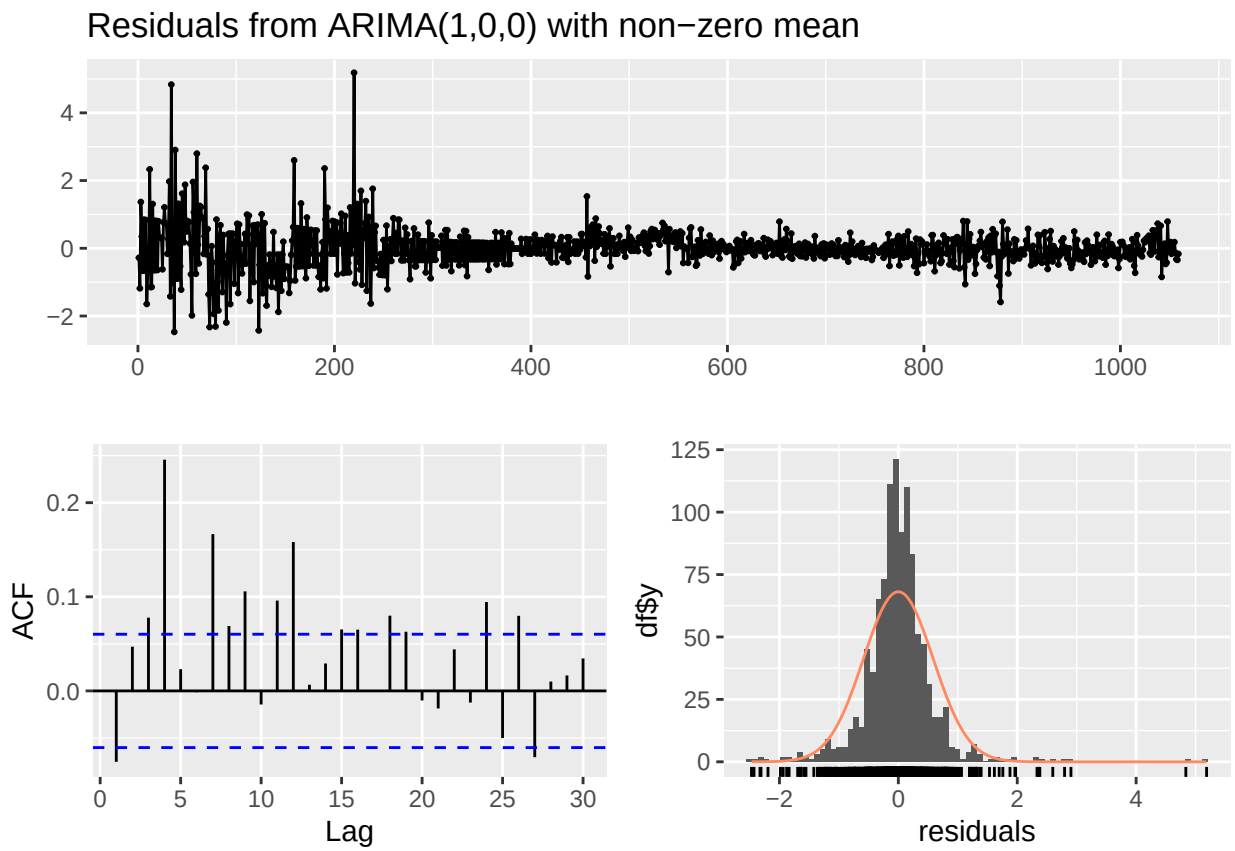
```
## Series: cpi_data$CPIAI
## ARIMA(1,1,1)
##
## Coefficients:
##          ar1      ma1
##       0.2382  -0.8586
## s.e.  0.0423   0.0254
##
## sigma^2 = 0.3277:  log likelihood = -910.51
## AIC=1827.02   AICc=1827.04   BIC=1841.91
##
## Training set error measures:
##                       ME      RMSE       MAE MPE MAPE      MASE         ACF1
## Training set 0.0005467244 0.5716334 0.3622792 NaN  Inf 0.8595134 0.0007668526
```

```
    # Fit ARIMA models (non-seasonal)
    fit_arima <- Arima(cpi_data$CPIAI, order = c(1, 0, 1))  # Adjust based on your data
    summary(fit_arima)
```

```
## Series: cpi_data$CPIAI
## ARIMA(1,0,1) with non-zero mean
##
## Coefficients:
##          ar1      ma1    mean
##       0.8847  -0.5899  0.3181
## s.e.  0.0268   0.0501  0.0618
##
## sigma^2 = 0.3241:  log likelihood = -904.83
## AIC=1817.67   AICc=1817.71   BIC=1837.53
##
## Training set error measures:
##                       ME      RMSE      MAE MPE MAPE      MASE      ACF1
## Training set 0.0007770963 0.5685077 0.366862 NaN  Inf 0.8703862 0.0588272
```

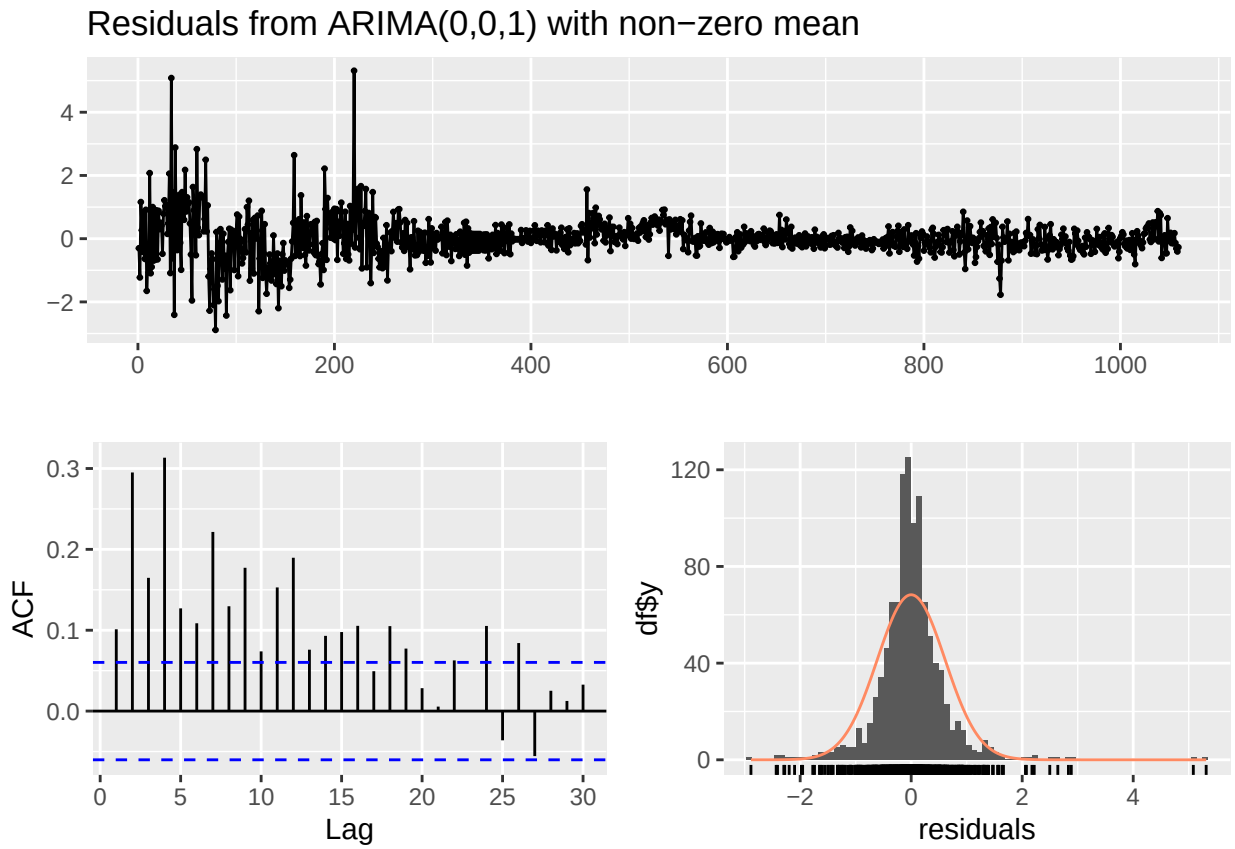
Step 4: Checking residuals:

```
# Check residuals for the AR model
checkresiduals(fit_ar)
```



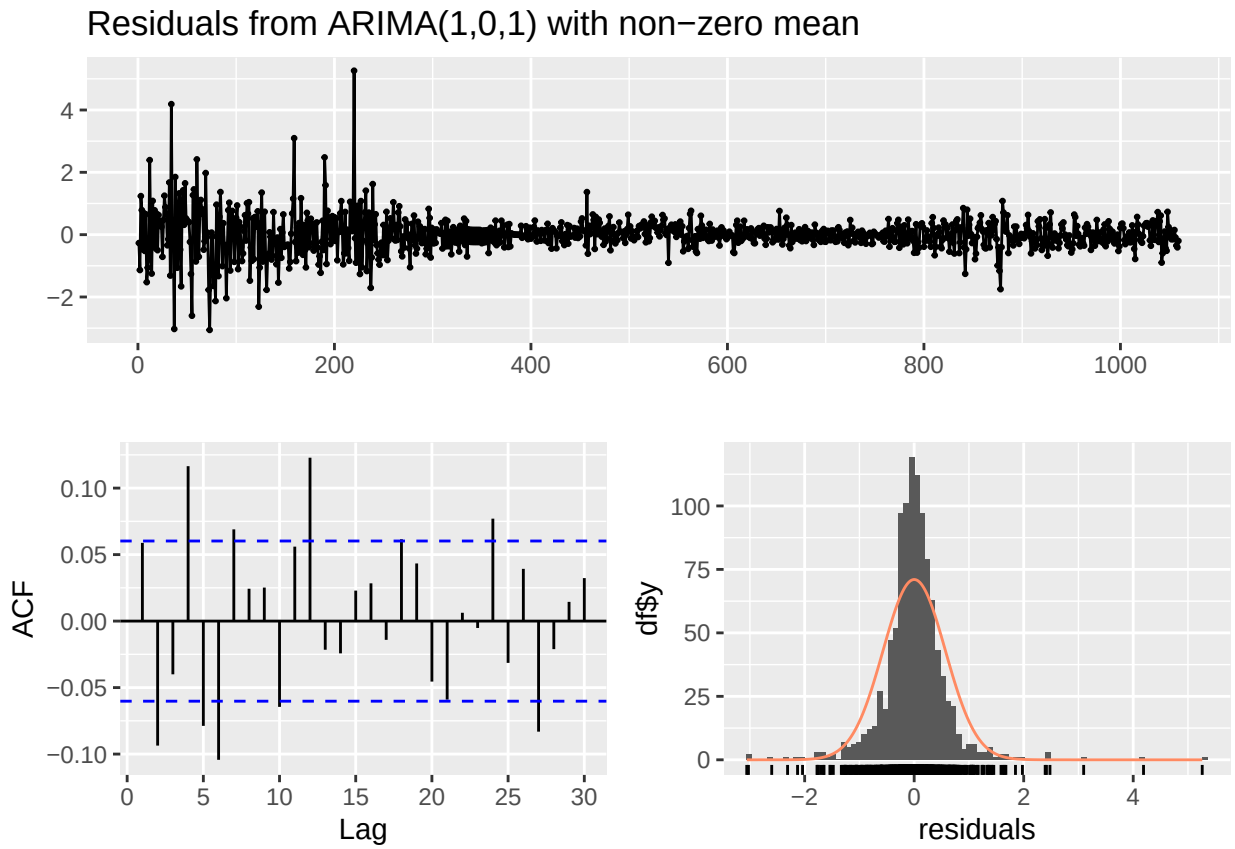

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,0) with non-zero mean
## Q* = 126.55, df = 9, p-value < 2.2e-16
##
## Model df: 1.   Total lags used: 10
```

```
# Check residuals for the MA model
checkresiduals(fit_ma)
```



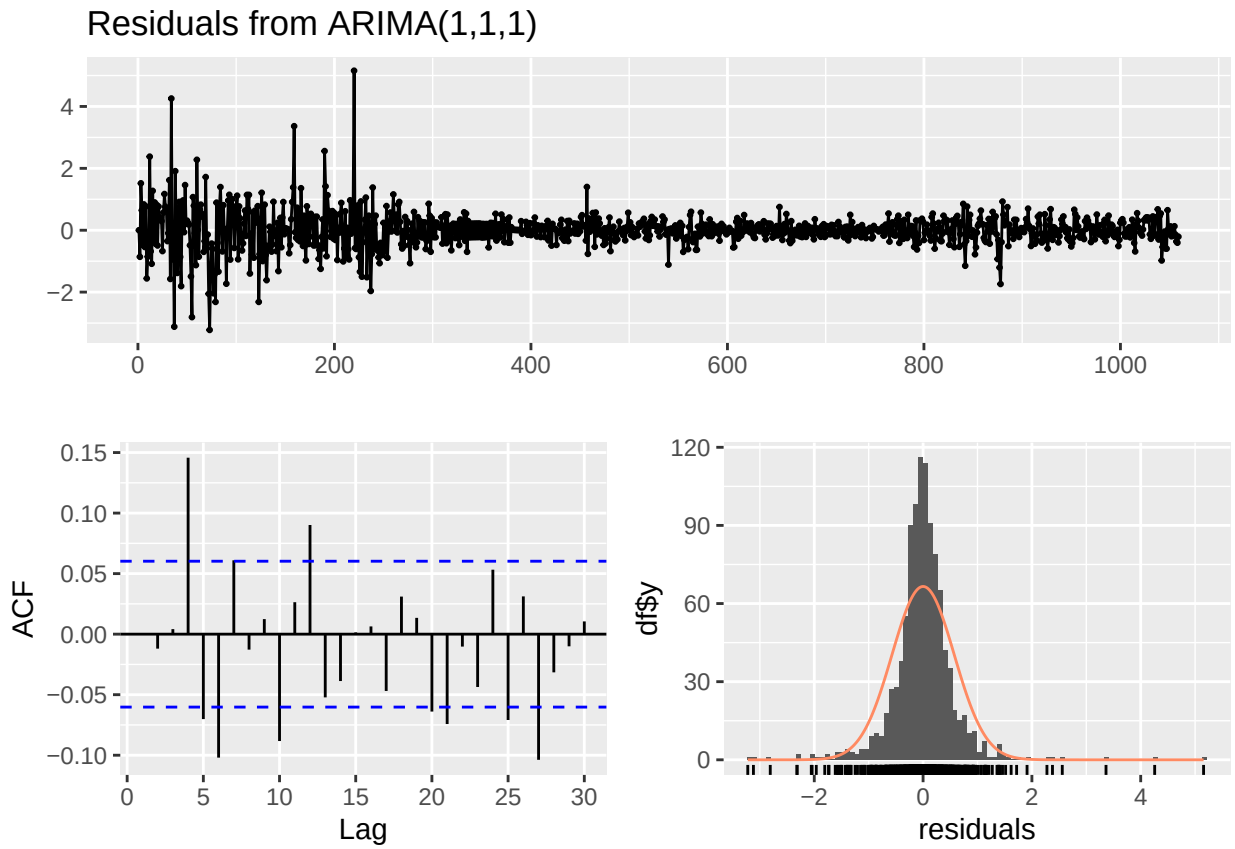

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,1) with non-zero mean
## Q* = 376.56, df = 9, p-value < 2.2e-16
##
## Model df: 1.   Total lags used: 10
```

```
# Check residuals for the ARMA model
checkresiduals(fit_arma)
```



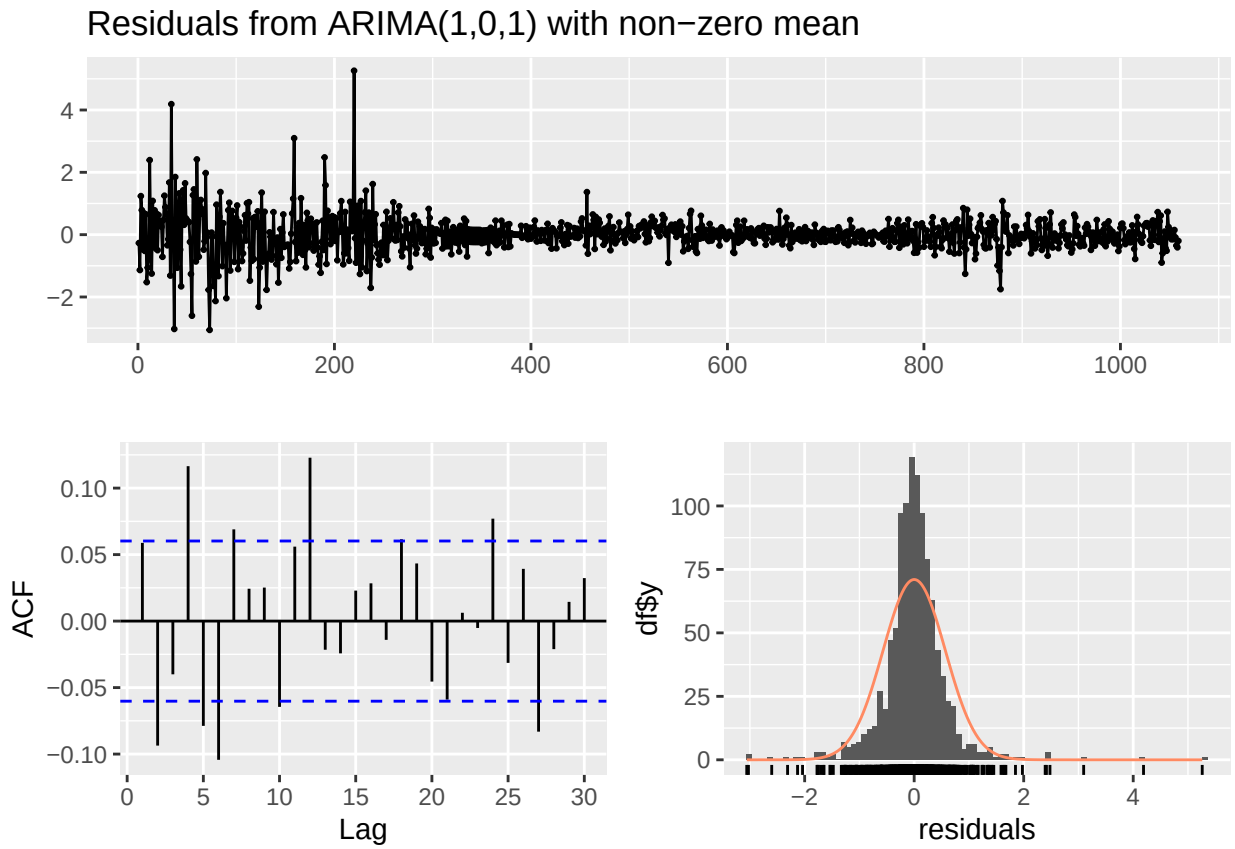

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,1) with non-zero mean
## Q* = 58.228, df = 8, p-value = 1.036e-09
##
## Model df: 2.   Total lags used: 10
```

```
# Check residuals for the SARIMA model
checkresiduals(fit_sarima)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,1,1)
## Q* = 51.795, df = 8, p-value = 1.844e-08
##
## Model df: 2.   Total lags used: 10
```

```
# Check residuals for the ARIMA model
checkresiduals(fit_arima)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(1,0,1) with non-zero mean
## Q* = 58.228, df = 8, p-value = 1.036e-09
##
## Model df: 2.   Total lags used: 10
```

Step 5: Box-Ljung test:

```
# Ljung-Box test for residual autocorrelation
Box.test(fit_ar$residuals, lag = 10, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_ar$residuals
## X-squared = 126.55, df = 10, p-value < 2.2e-16
```

```
Box.test(fit_ma$residuals, lag = 10, type = "Ljung-Box")
```

```
##
```

```
##  Box-Ljung test
##
## data:  fit_ma$residuals
## X-squared = 376.56, df = 10, p-value < 2.2e-16
```

```
Box.test(fit_arma$residuals, lag = 10, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_arma$residuals
## X-squared = 58.228, df = 10, p-value = 7.83e-09
```

```
Box.test(fit_sarima$residuals, lag = 10, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_sarima$residuals
## X-squared = 51.795, df = 10, p-value = 1.245e-07
```

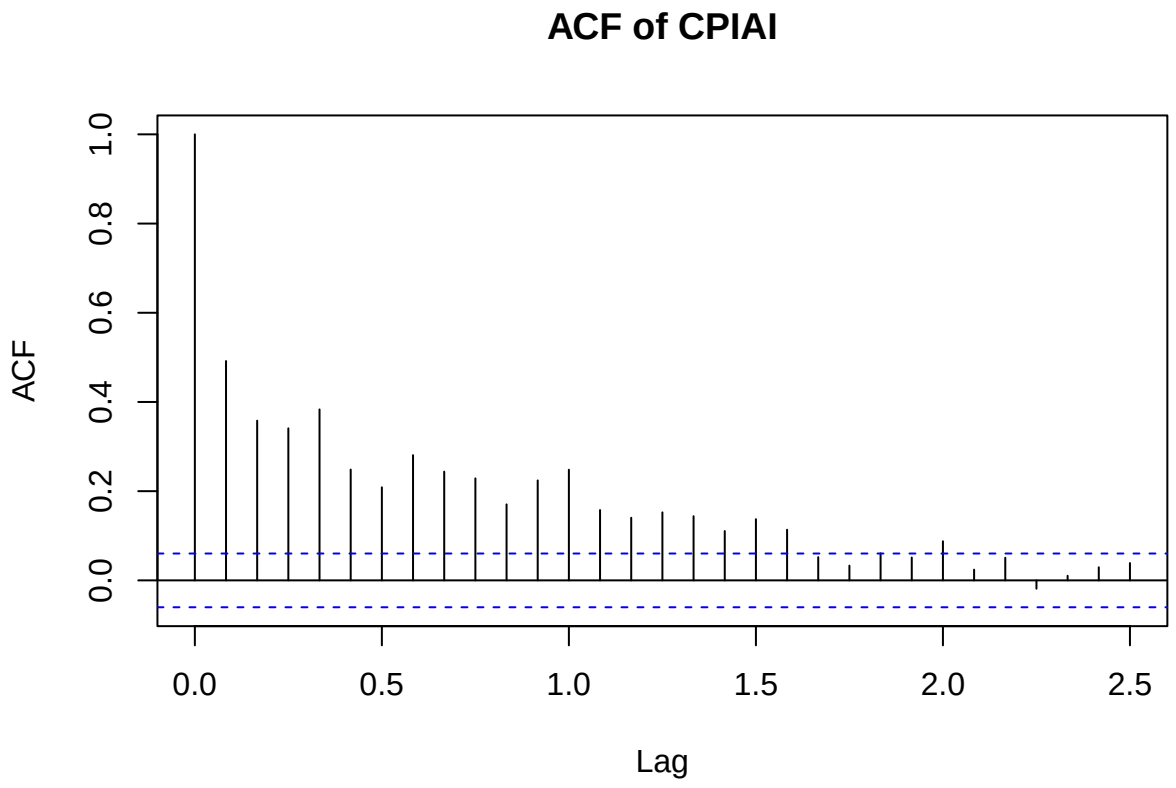
```
Box.test(fit_arima$residuals, lag = 10, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  fit_arima$residuals
## X-squared = 58.228, df = 10, p-value = 7.83e-09
```

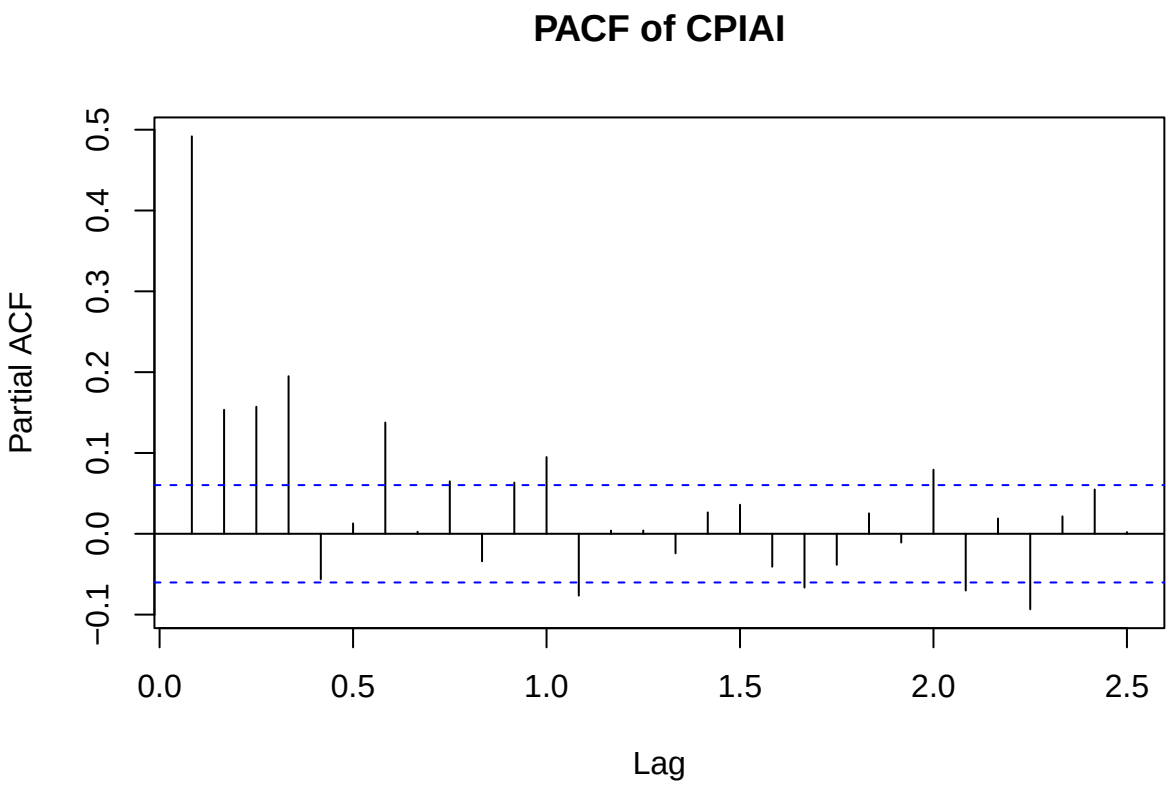
Step 6: ACF AND PACF PLOTS:

```
#convert the original CPIAI data to a time series object
cpiai_ts <- ts(cpi_data$CPIAI, frequency = 12, start = c(1948, 1))  # Adjust start year and frequency i

#ACF and PACF for the original data
acf(cpiai_ts, main = "ACF of CPIAI")
```




```
pacf(cpiai_ts, main = "PACF of CPIAI")
```



Step 7: Forecasting

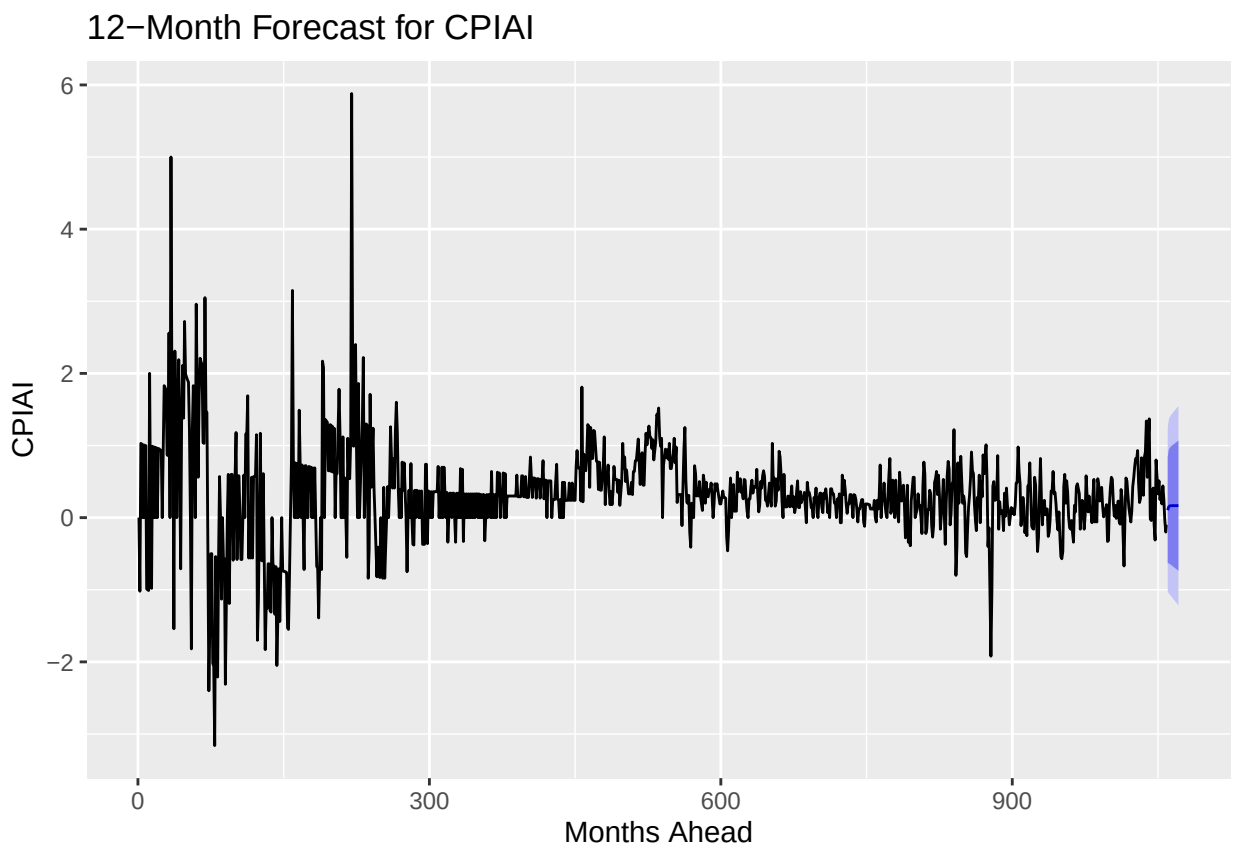
```
# Forecasting the next 12 months
# fitting SARIMA model
fit_sarima <- Arima(cpi_data$CPIAI, order = c(1, 1, 1), seasonal = c(1, 1, 1))  # SARIMA model

# Check the fitted SARIMA model
summary(fit_sarima)
```

```
## Series: cpi_data$CPIAI
## ARIMA(1,1,1)
##
## Coefficients:
##          ar1      ma1
##       0.2382  -0.8586
## s.e.  0.0423   0.0254
##
## sigma^2 = 0.3277:  log likelihood = -910.51
## AIC=1827.02   AICc=1827.04   BIC=1841.91
##
## Training set error measures:
##                       ME      RMSE       MAE MPE MAPE      MASE         ACF1
## Training set 0.0005467244 0.5716334 0.3622792 NaN  Inf 0.8595134 0.0007668526
```

```r
# Forecast using the SARIMA model
cpi_forecast <- forecast(fit_sarima, h = 12)

# Plot the forecast
autoplot(cpi_forecast) +
  ggtitle("12-Month Forecast for CPIAI") +
  xlab("Months Ahead") +
  ylab("CPIAI")
```



Step 9: Final Conclusion :

In this analysis, I applied various models—AR, MA, ARMA, and SARIMA—to predict CPIAI values. Among these, the **SARIMA(1,1,1) model**, which captures both trend and seasonality, proved to be the most accurate, as evidenced by its strong performance according to AIC and BIC criteria. The Ljung-Box test showed that the residuals were white noise, confirming the model's effectiveness in capturing the underlying data patterns. By forecasting the next 12 months,can anticipate future CPIAI trends, showcasing the power of time series forecasting for economic insights.